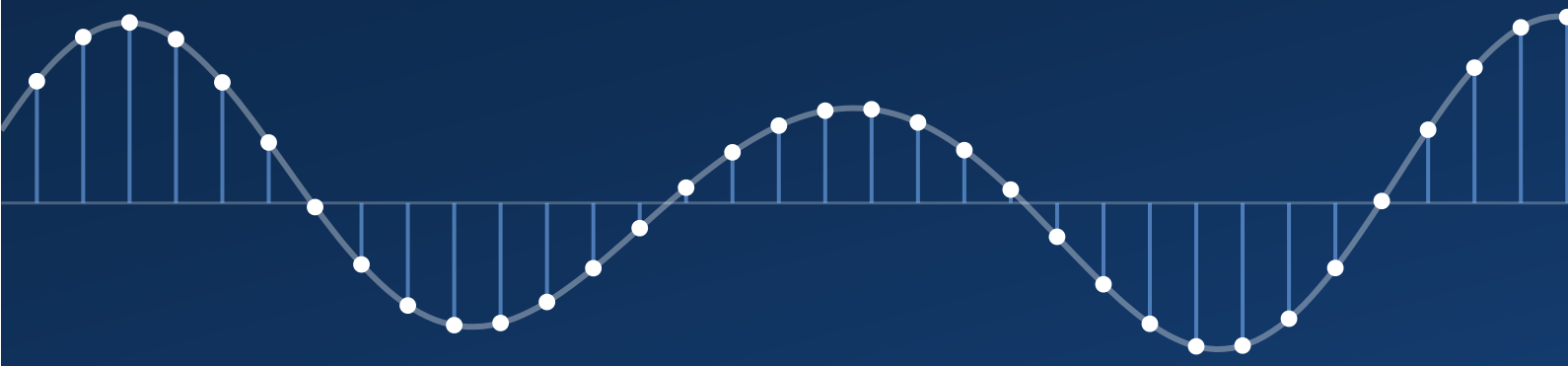
DIGITAL SIGNAL PROCESSING

CHAPTER 01

النظري والمسائل

كل الأسئلة النظري من محاضرات 2 و 3 و 4، وكل المسائل من المحاضرات والامتحانات القديمة، بإجابات نموذجية بجُمل الدكتور وطريقته في الحل.

Signals & Systems | Frequency | A/D & D/A | Sampling & Aliasing



22	16	+20
سؤال نظري	مسألة محلولة	امتحان قديم متراجع

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إزاي تستخدم الملف

النظري بجُمل الدكتور من المحاضرة، مترتب (تعريف ← شرح ← معادلة ← رسمة) عشان يبان إنك فاهم. والمسائل خطوة بخطوة بنفس طريقة الدكتور في أمثلة المحاضرة (Examples 1.1 لحد 1.4). والسطر البرتقالي فيه الامتحانات اللي السؤال جه فيها قبل كده.

★ الأسئلة النظري اللي بتتكرر أكتر (ابدأ بيها)

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SECTION 01

محاضرة 2

Signals, Systems & Classification

1. Define the following items: Signal, System, and Signal Processing (give an example for each).
2. Draw the block diagram of (a) an analog signal processor and (b) a digital signal processor.
3. What are the advantages of digital signal processing over analog signal processing? What is its…
4. Explain, with drawings, the different classifications of signals.
5. Compare between deterministic and random signals.
6. What is a digital signal? Why must a signal be digital to be processed digitally?
7. What is quantization? Compare between truncation and rounding with an example.

Q1 Define the following items: Signal, System, and Signal Processing (give an example for each).

ANSWER

- **Signal:** is defined as any physical quantity that varies with time, space, or any other independent variable(s). Mathematically, we describe a signal as a function of one or more independent variables.
  Examples: $s1(t) = 5t$ (varies linearly with t), $s2(t) = 20t^2$ (varies quadratically with t), and $s(x, y) = 3x + 2xy + 10y^2$ (two independent variables, e.g. the spatial coordinates in a plane). A speech signal cannot be described by such an expression; a segment of it is represented as a sum of several sinusoids of different amplitudes, frequencies and phases.
- **System:** is defined as a physical device that performs an operation on a signal.
  Example: a filter used to reduce the noise and interference corrupting a desired information. If the operation is linear, the system is called linear; if it is nonlinear, the system is nonlinear.
- **Signal processing:** when we pass a signal through a system, as in filtering, we say that we have processed the signal.
  Example: filtering the noise and interference from the desired signal. Digital processing may be performed by digital hardware (logic circuits), by software, or by a combination of both.

Quiz 1 (Section) Midterm OLD (Model-C) Sheet 1 (1.1) جه في

Q2 Draw the block diagram of (a) an analog signal processor and (b) a digital signal processor.

ANSWER

There are two types of signal processing systems (Sec. 1.2):

1. **Analog signal processor (Fig. 1.1):** the analog input signal is processed directly by an analog signal processor to give an analog output signal.
2. **Digital signal processor (Fig. 1.2):** provides an alternative method for processing the analog signal. The **A/D converter** converts the analog input into a digital signal (a sequence of numbers), the **digital signal processor** performs the required operations on it, and the **D/A converter** converts the result back into an analog output signal.





Sheet 1 (1.2) Revision Sheet 1 جه في

Q3 What are the advantages of digital signal processing over analog signal processing? What is its main limitation?

ANSWER

1. **Flexibility:** a digital programmable system allows reconfiguring the operations simply by changing the program; an analog system needs hardware redesign.
2. **Accuracy:** a digital system provides much better control of accuracy requirements; tolerances in analog components make accuracy hard to control.
3. **Storage:** digital signals are easily stored on magnetic media without deterioration, so they are transportable and can be processed off-line.
4. Allows the implementation of more **sophisticated** signal processing algorithms.
5. **Precise mathematical operations** are very difficult in analog form but routine on a digital computer using software.
6. In some cases a digital implementation is **cheaper** than its analog counterpart.

Limitation: the speed of operation of A/D converters and digital signal processors; signals with extremely wide bandwidths may be beyond the state of the art of digital hardware.

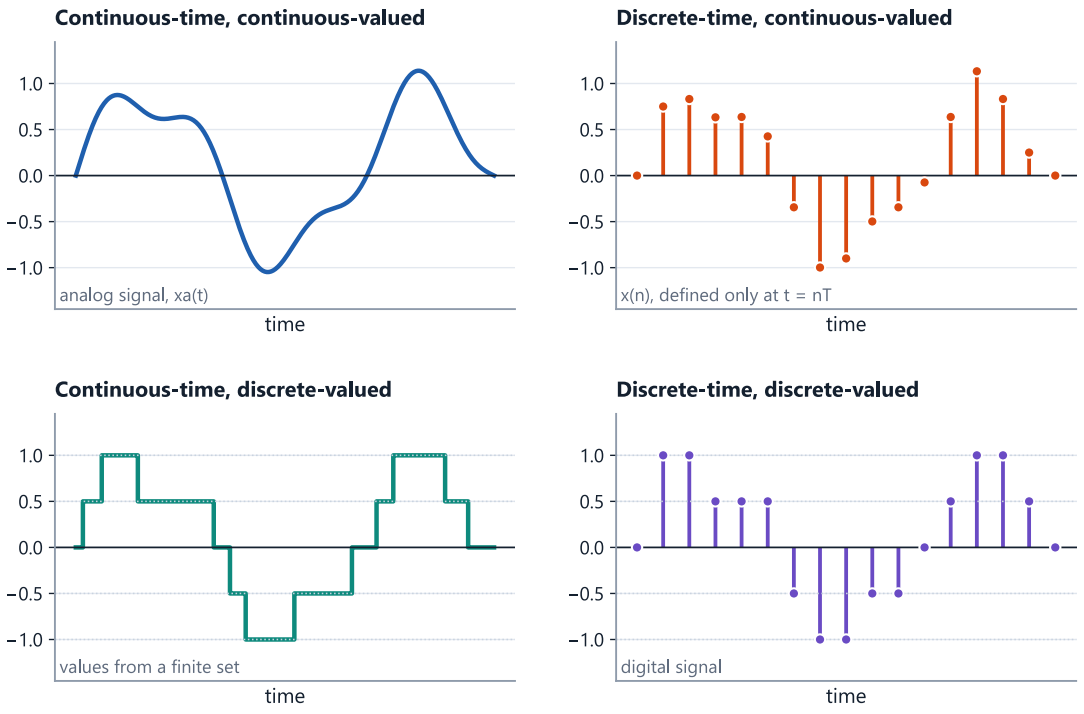
Midterm OLD جه في

Q4 Explain, with drawings, the different classifications of signals. متكرر في الامتحانات

ANSWER

1. **Continuous-time vs. discrete-time:** continuous-time (analog) signals are defined for every value of time and take on values in the continuous interval (a, b). Discrete-time signals are defined only at certain specific values of time.
2. **Continuous-valued vs. discrete-valued:** a continuous-valued signal takes on all possible values on a finite or an infinite range. A discrete-valued signal takes on values from a finite set of possible values.
3. **Deterministic vs. random:** a deterministic signal can be uniquely described by an explicit mathematical expression, a table of data, or a well-defined rule. Otherwise, the signal is random.

4. **Multichannel vs. multidimensional:** a multichannel signal is generated by multiple sources or sensors and represented in vector form, $S3(t) = [s1(t)\ s2(t)\ s3(t)]^T$. A signal is M-dimensional if its value is a function of M independent variables.



نظري الميد | Sheet 1 (1.3) | Revision Sheet 1 | Midterm 8-12-2020 (Q2-a) | جه في

Q5 Compare between deterministic and random signals.

ANSWER

- **Deterministic:** any signal that can be uniquely described by an explicit mathematical expression, a table of data, or a well-defined rule. All past, present, and future values are known precisely, without any uncertainty.
- **Random:** otherwise, we refer to these signals as random.

Midterm OLD (Model-A 2013/2014) | جه في

Q6 What is a digital signal? Why must a signal be digital to be processed digitally?

ANSWER

- A **digital signal** is discrete in time and its values are discrete (Fig. 1.4: a digital signal with four different amplitude values).
- In order for a signal to be processed digitally, it must be discrete in time and its values must be discrete (i.e., it must be a digital signal).

Q7 **What is quantization? Compare between truncation and rounding with an example.**

ANSWER

- **Quantization:** the process of converting a continuous-valued signal into a discrete-valued signal; it is basically an approximation process, accomplished by rounding or truncation.
- **Example:** if the allowable values are the integers 0 through 15, the value 8.58 is approximated by **8** by truncation, or by **9** by rounding to the nearest integer.

SECTION 02

محاضرة 3

Concept of Frequency

1. In detail, what are the main characteristics of the continuous-time sinusoidal signals?
2. In detail, what are the main characteristics of the discrete-time sinusoidal signals?
3. Show that a discrete-time sinusoid is periodic only if its frequency F is a rational number. Ho…
4. Prove that discrete-time sinusoids whose frequencies are separated by an integer multiple of 2π…
5. What are harmonically related complex exponentials? Compare between the continuous-time and dis…

Q8 In detail, what are the main characteristics of the continuous-time sinusoidal signals?

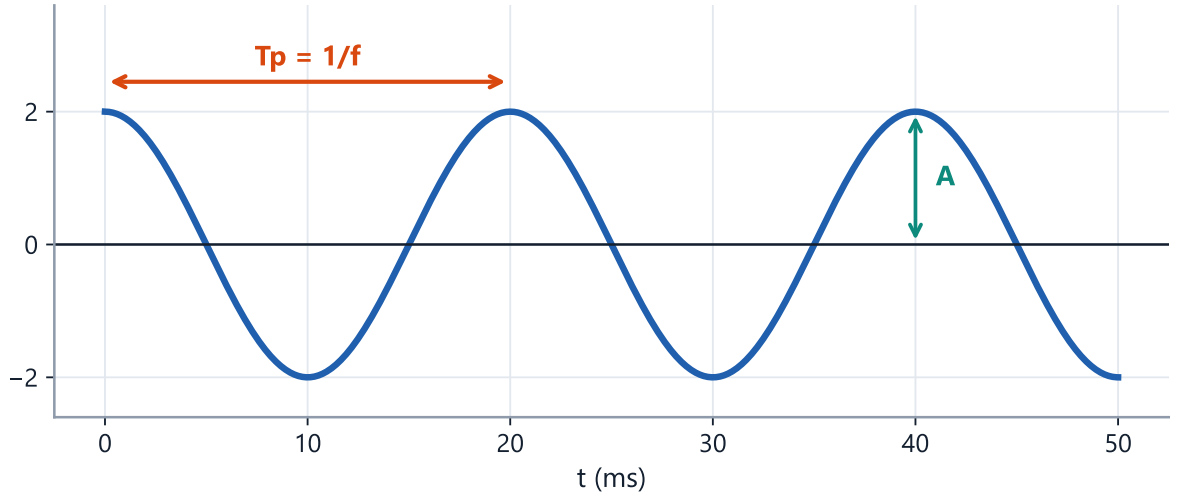
متكرر في الامتحانات

ANSWER

$x_a(t) = A\cos(\omega t + \theta) = A\cos(2\pi f t + \theta)$, where A is the amplitude, $\omega = 2\pi f$ the frequency in rad/s, f in Hz, and θ the phase in radians.

1. **(a)** For every fixed value of the frequency f, $x_a(t)$ is **periodic**: $x_a(t + T_p) = x_a(t)$, where $T_p = 1/f$ is the fundamental period.
2. **(b)** Continuous-time sinusoidal signals with distinct (different) frequencies are themselves **distinct**.
3. **(c)** Increasing the frequency f results in an **increase in the rate of oscillation** of the signal; f can be increased without limit.

Continuous-time sinusoid: periodic for every f, Tp = 1/f



Midterm OLD | Quiz 1 Summer 26 (Q1-b) | Final 9-6-2026 (Q1-a) | Final 20-1-2025 (Q3-a) جه في

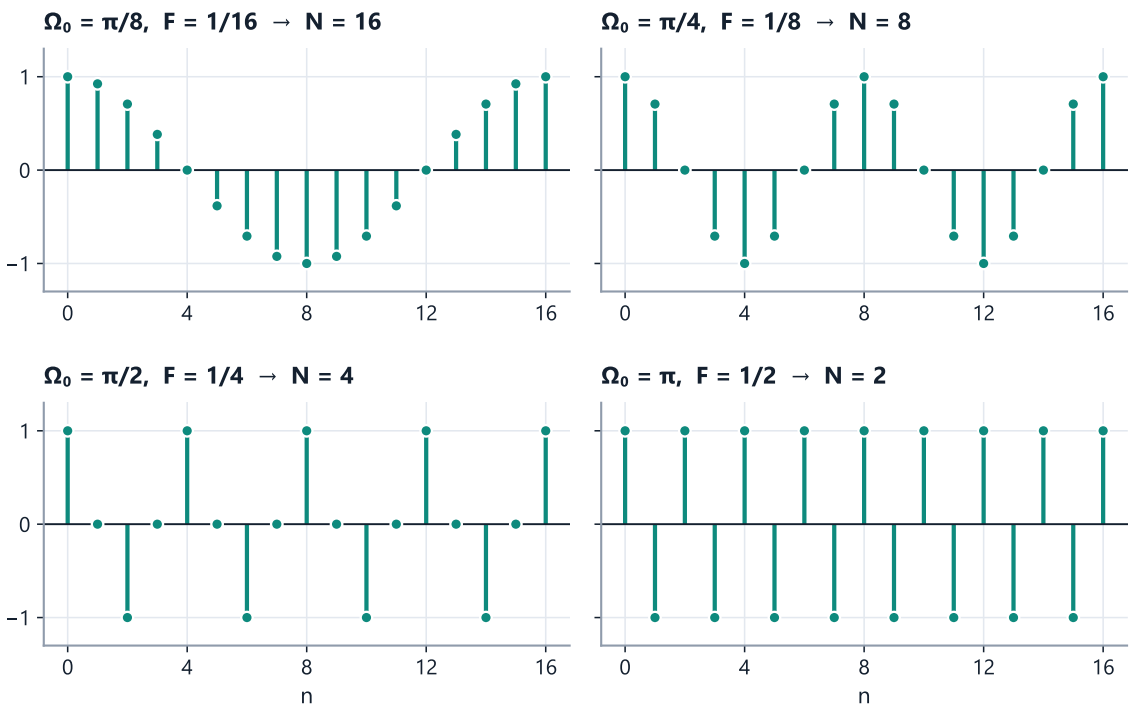
Q9 In detail, what are the main characteristics of the discrete-time sinusoidal signals?

متكرر في الامتحانات

ANSWER

$x(n) = A\cos(\Omega n + \theta) = A\cos(2\pi F n + \theta)$, where n is the sample number, Ω is in radians/sample and F in cycles/sample.

1. **(a)** A discrete-time sinusoid is **periodic only if its frequency F is a rational number** ($F_0 = k/N$).
2. **(b)** Discrete-time sinusoids whose frequencies are separated by an integer multiple of 2π are **identical**. Frequencies in $-\pi \le \Omega \le \pi$ ($-1/2 \le F \le 1/2$) are unique; all frequencies $|\Omega| > \pi$ are **aliases**.
3. **(c)** The **highest rate of oscillation** is attained when $\Omega = \pi$ (or $\Omega = -\pi$), i.e. $F = 1/2$ (or $F = -1/2$).



Final 23-12-2017 Final 2-1-2017 Final 12-1-2019 Final 7-6-2023 (Q4-a) Final 11-6-2023 جه في

نظري فينال Midterm OLD Midterm 31-7-2023 Midterm 26-7-2026

Q10 **Show that a discrete-time sinusoid is periodic only if its frequency F is a rational number. How do you find its fundamental period?**

ANSWER

By definition, x(n) is periodic with period N (N > 0) if and only if x(n + N) = x(n) for all n (1.15). For a sinusoid with frequency F_0 to be periodic:

$$\cos[2\pi F_0(N + n) + \theta] = \cos(2\pi F_0 n + \theta)$$

This relation is true if and only if there exists an integer k such that $2\pi F_0 N = 2k\pi$, or equivalently:

$$F_0 = k / N \quad (1.16)$$

So F_0 must be rational. To find the fundamental period, cancel common factors so that k and N are relatively prime; then the fundamental period is N.

Notebook / Section Midterm 1 Model-2 (Q2-b) جه في

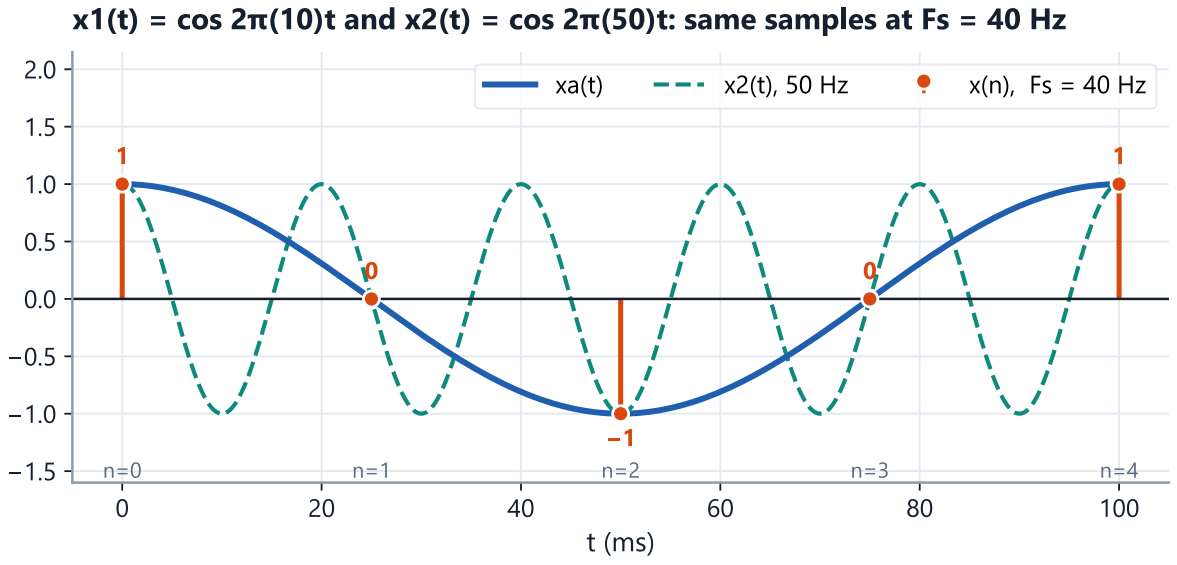
Q11 Prove that discrete-time sinusoids whose frequencies are separated by an integer multiple of 2π are identical. What is an alias?

ANSWER

$$\cos[(\Omega_0 + 2\pi)n + \theta] = \cos(\Omega_0 n + 2\pi n + \theta) = \cos(\Omega_0 n + \theta) \quad (1.17)$$

As a result, all sinusoidal sequences with $\Omega k = \Omega_0 + 2k\pi$, $-\pi \le \Omega_0 \le \pi$, are indistinguishable (identical).

Alias: any sequence from a sinusoid with $|\Omega| > \pi$ ($|F| > 1/2$) is identical to a sequence obtained from a sinusoid with $|\Omega| < \pi$; it is called an alias of the corresponding sinusoid.



Q12 What are harmonically related complex exponentials? Compare between the continuous-time and discrete-time cases.

ANSWER

Sets of periodic complex exponentials with fundamental frequencies that are multiples of a single positive frequency.

	Continuous time	Discrete time
Basic signals	$s_k(t) = e^{jk\Omega_0 t}$, k = 0, ±1, ±2, …	$s_k(n) = e^{j2\pi kn/N}$, k = 0, 1, …, N − 1
Distinct members	all distinct (infinite)	only N distinct: $s_{k+N}(n) = s_k(n)$
Common period	$Tp = 1/F_0$	N samples
Fourier series	$xa(t) = \Sigma\, c_k\, e^{jk\Omega_0 t}$	$x(n) = \Sigma_{k=0..N-1}\, c_k\, e^{j2\pi kn/N}$

SECTION 03

محاضرة 4

A/D and D/A Conversion

1. Explain (with a block diagram) the basic three operations performed in the A/D converter.
2. In A/D, explain in detail the sampling process of analog signals.
3. Find the relationship between the frequency of the analog signal f, the frequency of the discre…
4. In sampling process, derive the sampling condition Fs ≥ 2fmax (avoiding aliasing).
5. State and explain the sampling theorem. Define the Nyquist rate.
6. In A/D, explain in detail the quantization process of continuous-amplitude signals.
7. In A/D conversion, explain in detail the coding process.
8. Why is the sampling process invertible while the quantization process is not?
9. Explain the D/A conversion process. Compare between ideal interpolation, zero-order hold and li…
10. Check (✓) for the right statements and (✗) for the wrong statements and correct the mistakes.

Q13 Explain (with a block diagram) the basic three operations performed in the A/D converter.

متكرر في الامتحانات

ANSWER

To process analog signals by digital means, it is first necessary to convert them into a sequence of numbers having finite precision. This is A/D conversion, a three-step process (Fig. 1.9):



1. **Sampling:** the conversion of a continuous-time signal into a discrete-time signal obtained by taking "samples" of the continuous-time signal at discrete-time instants. If xa(t) is the input to the sampler, the output is **x(n) = xa(nT)**, where T is the sampling interval.
2. **Quantization:** the conversion of a discrete-time continuous-valued signal into a discrete-time, discrete-valued (digital) signal. The value of each sample is selected from a finite set of possible values. The difference $x_q(n) - x(n)$ is the quantization error.
3. **Coding:** each discrete value $x_q(n)$ is represented by a **b-bit binary sequence**.

Sampling does not result in a loss of information if the signal bandwidth is finite (**invertible**), while quantization is a **noninvertible** process that results in signal distortion.

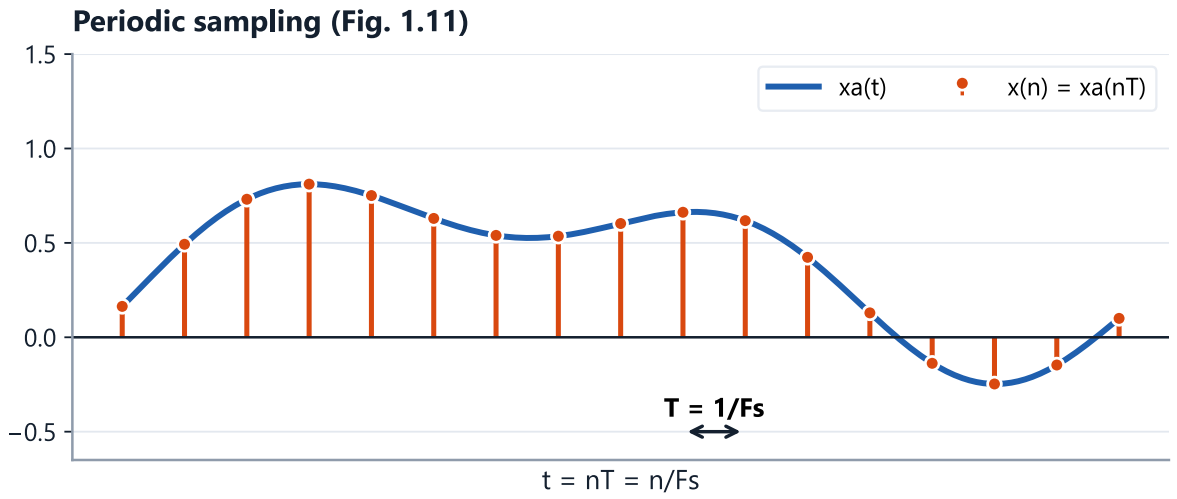
نظري الميد | Sheet 1 (1.4) | Midterm OLD (Model-A 2012/2013) | Revision Sheet 1 | جه في

Q14 In A/D, explain in detail the sampling process of analog signals.

متكرر في الامتحانات

ANSWER

- We use periodic (uniform) sampling, described by the relation **x(n) = xa(nT)**, $-\infty < n < \infty$ (1.29), where x(n) is obtained by taking samples of xa(t) every T seconds (Fig. 1.11).
- The time interval T between successive samples is the **sampling period**; its reciprocal **Fs = 1/T** is the sampling rate (samples/s) or sampling frequency (Hz).
- Periodic sampling relates the time variables t and n: **t = nT = n/Fs** (1.30). As a consequence, the frequencies are related by F = f/Fs (1.33).
- Sampling does not result in a loss of information if the bandwidth is finite and $Fs > 2f_{max}$.



Midterm OLD | Midterm 1 Model-2 (Q1-a) | Midterm 8-12-2020 (Q1-a) | Midterm 28-11-2021 (Q1-a) جه في

Q15 **Find the relationship between the frequency of the analog signal f, the frequency of the discrete-time signal F, and the sampling frequency Fs.** متكرر في الامتحانات

ANSWER

Consider $x_a(t) = A\cos(2\pi ft + \theta)$. Sampling at $F_s = 1/T$ ($t = nT = n/F_s$):

$$x(n) = x_a(nT) = A\cos(2\pi fnT + \theta) = A\cos(2\pi nf/F_s + \theta) \quad (1.32)$$

Comparing with $x(n) = A\cos(2\pi Fn + \theta)$ (1.14):

$$\mathbf{F = f / F_s} \ (1.33) \quad \text{or} \quad \mathbf{\Omega = \omega T} \ (1.34)$$

Final 23-12-2017 (Q2-a) | Final 2-1-2017 (Q2-a) | Final 12-1-2019 (Q2-a) | Final 18-1-2024 (Q1-a) جه في
Final 8-9-2021 | Midterm 2 (Q2-a) | Midterm 12-11-2025 (Q1-a) | Final 28-12-2019 (Q1-a)

Q16 **In sampling process, derive the sampling condition $F_s \geq 2f_{max}$ (avoiding aliasing).** متكرر في الامتحانات

ANSWER

Since the discrete-time frequency must satisfy $-1/2 \leq F \leq 1/2$ and $F = f/F_s$:

$$-1/2 \leq f/F_s \leq 1/2 \ \rightarrow \ -F_s/2 \leq f \leq F_s/2 \quad (1.37)$$
$$f_{max} = F_s/2 \ \rightarrow \ \mathbf{F_s \geq 2f_{max}}$$

Thus to avoid the problem of aliasing, Fs is selected so that Fs > $2f_{max}$ (1.47).

جه في Midterm 26-7-2026 (Q2-a) Final 27-6-2021 (Q1-a) Final 20-1-2025 (Q4-a) Quiz 1 Summer 26 (Q1-a) Midterm 2 (Q1-a) Midterm 31-7-2023 (Q2-a)

Q17 State and explain the sampling theorem. Define the Nyquist rate.

ANSWER

- To select Fs we must have information about the frequency content of the signal, e.g. f_{max} = 3000 Hz for speech and 5 MHz for television signals. The signal may be passed through a filter that severely attenuates components above f_{max}.
- To avoid aliasing, Fs is selected so that **Fs > $2f_{max}$** (1.47). Then every component $|f_i| < f_{max}$ is mapped into $-1/2 \le F_i = f_i/Fs \le 1/2$ (1.48).

Sampling theorem: if the highest frequency contained in an analog signal xa(t) is f_{max} = B and the signal is sampled at a rate Fs > $2f_{max}$ = 2B, then xa(t) can be exactly recovered from its sample values using the interpolation function:

$$g(t) = \sin(2\pi Bt) / (2\pi Bt) \quad (1.50) \qquad x_a(t) = \Sigma\, x_a(n/Fs)\, g(t - n/Fs) \quad (1.51)$$

- The sampling rate **$f_N = 2B = 2f_{max}$** is called the **Nyquist rate**.
- Because of the complexity and the infinite number of samples required, these reconstruction formulas are primarily of theoretical interest.

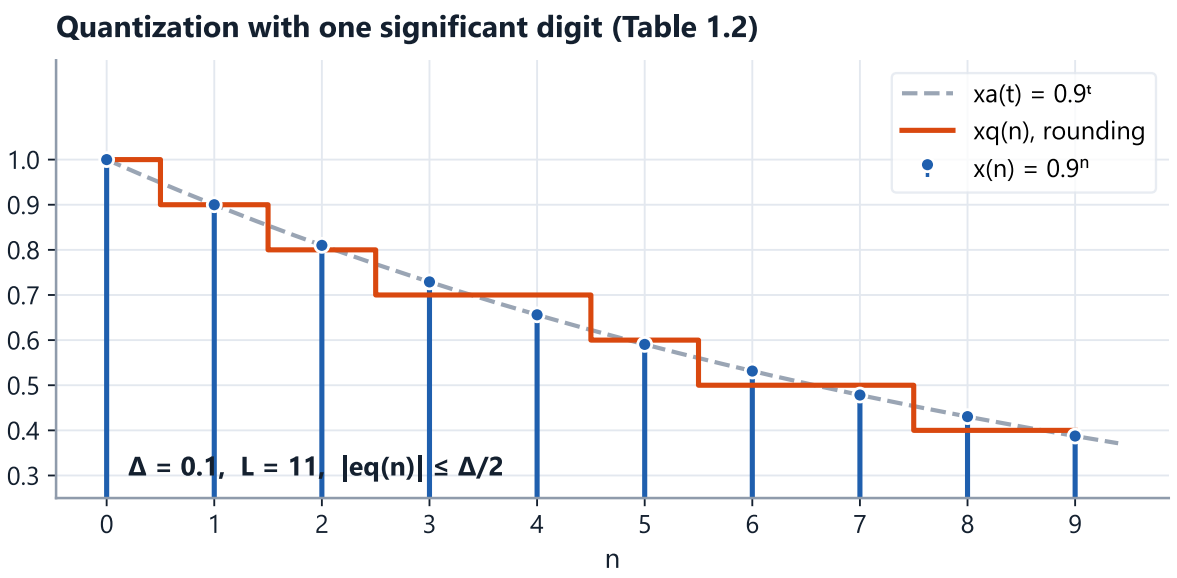
جه في Sheet 1 (1.5) نظري المبد Eng.Khalaf نظري

Q18 In A/D, explain in detail the quantization process of continuous-amplitude signals.

متكرر في الامتحانات

ANSWER

- **Quantization:** the process of converting a discrete-time continuous-amplitude signal into a digital signal by expressing each sample value as a finite number of digits.
- **Quantization error:** eq(n) = xq(n) − x(n) (1.53), where xq(n) = Q[x(n)].
- The values allowed are the **quantization levels**; the distance between two successive levels is the **step size Δ**.
- In rounding: $-\Delta/2 \le eq(n) \le \Delta/2$ (1.54), and $\Delta = (x_{max} - x_{min})/(L - 1)$ (1.55).
- Increasing the number of levels L decreases Δ, so the quantization error decreases and the accuracy increases.



Final 8-9-2021 (Q1-a) Final 4-3-2021 (Q1-a) Final 17-8-2026 (Q1-a) Final 7-6-2023 (Q1-a) جه في

نظري فينال Midterm OLD Quiz 1 (Section) Midterm 2 (Q1-b)

Q19 In A/D conversion, explain in detail the coding process.

ANSWER

- The coding process assigns a unique binary number to each quantization level.
- With L levels we need at least L binary numbers; with b bits we can create 2^b numbers.

$$2^b \geq L \rightarrow b \geq \log_2 L$$

Book exam Midterm 2015 (Model-A) جه في

Q20 Why is the sampling process invertible while the quantization process is not?

ANSWER

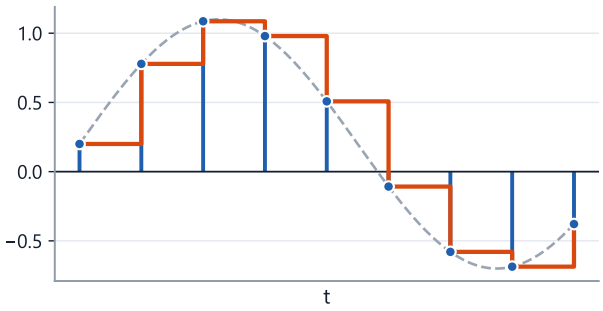
- **Sampling** does not result in a loss of information, nor introduce distortion, if the signal bandwidth is finite → **invertible (reversible)**.
- **Quantization** is a **noninvertible (irreversible)** process that results in signal distortion.

Q21 Explain the D/A conversion process. Compare between ideal interpolation, zero-order hold and linear interpolation.

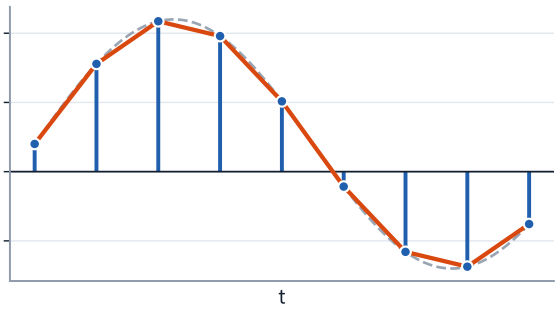
ANSWER

- The task of a D/A converter is to **interpolate between samples**. The ideal interpolation (sampling theorem) is too complicated and impractical.
- **Zero-order hold:** the simplest; holds constant the value of one sample until the next one is received.
- **Linear interpolation:** connects successive samples with straight-line segments (better).

Zero-order hold (Fig. 1.13)



Linear interpolation (Fig. 1.14)



Q22 Check (✓) for the right statements and (✗) for the wrong statements and correct the mistakes.

متكرر في الامتحانات

ANSWER

1. In quantization process, when the quantization step (Δ) increases, the accuracy of the quantizer increases. **(✗)**
   Correct: when Δ **decreases**, the accuracy increases.
2. Nyquist frequency (F_N) is the minimum sampling frequency. **(✓)**
3. To reconstruct x(t) from x(n), the sampling rate Fs must be less than the frequency of x(t), f_0. **(✗)**
   Correct: Fs must be **greater than $2f_0$**.

Quiz 1 Summer 26 (Q3) Final 27-6-2021 (Q4) Final 23-12-2017 (Q5-b) Final 12-1-2019 (Q5) جه في

SECTION 04

مسائل المحاضرات

Examples

1. Example 1.1 — Sampling & aliasing
2. Example 1.2 — Nyquist rate
3. Example 1.3 — Aliasing with three components
4. Example 1.4 — The aliasing concept
5. Quantization example (Table 1.2)
6. Periodicity examples (Lec 3)

P1 Example 1.1 — Sampling & aliasing

Consider the analog signal $x_a(t) = 3 \cos 100\pi t$.

a) Determine the minimum sampling rate required to avoid aliasing.

b) The signal is sampled at Fs = 200 Hz. What is the discrete-time signal?

c) The signal is sampled at Fs = 75 Hz. What is the discrete-time signal?

d) What is the frequency $0 < f < Fs/2$ of a sinusoid that yields samples identical to those in (c)?

SOLUTION

a) The frequency of the analog signal is f = 50 Hz. Hence the minimum sampling rate required to avoid aliasing is Fs = 100 Hz.

b) If the signal is sampled at Fs = 200 Hz, the discrete-time signal is

$$x(n) = 3 \cos(100\pi/200)n = 3 \cos(\pi/2)n$$

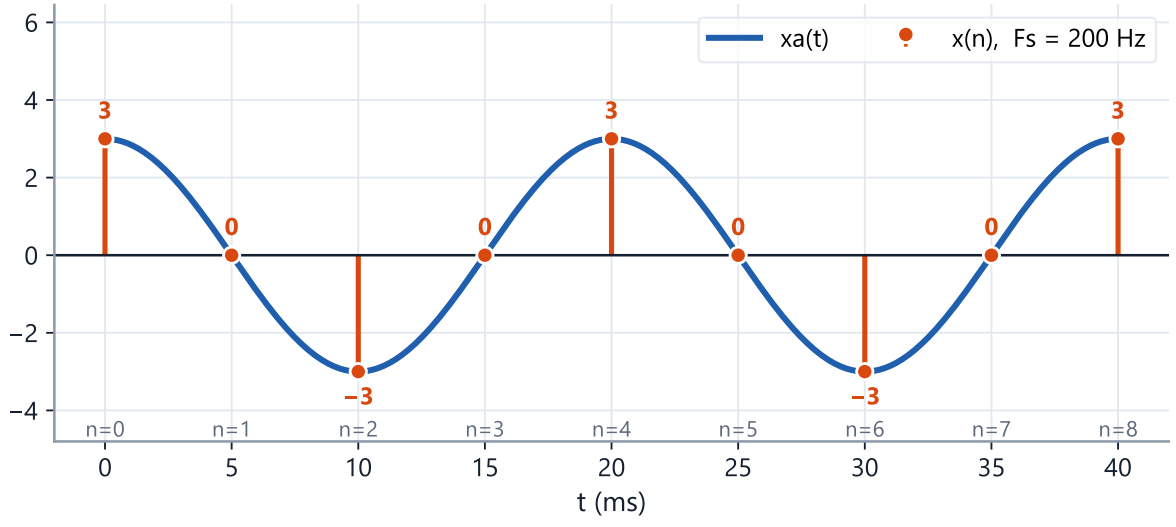
c) If the signal is sampled at Fs = 75 Hz, the discrete-time signal is

$$x(n) = 3 \cos(100\pi/75)n = 3 \cos(4\pi/3)n = 3 \cos(2\pi - 2\pi/3)n = 3 \cos(2\pi/3)n$$

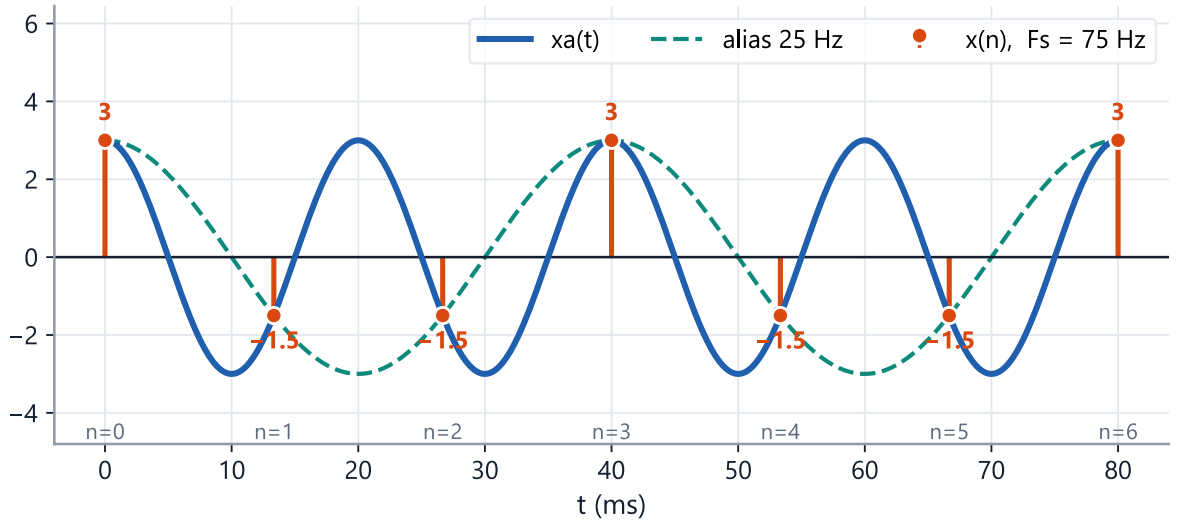
d) For the sampling rate of Fs = 75 Hz, we have $f = F \cdot Fs = 75F$. The frequency of the sinusoid in part (c) is F = 1/3. Hence f = 25 Hz.

Clearly, the sinusoidal signal $y_a(t) = 3 \cos 2\pi ft = 3 \cos 50\pi t$ sampled at Fs = 75 samples/s yields identical samples. Hence f = 50 Hz is an alias of f = 25 Hz for the sampling rate Fs = 75 Hz.

Example 1.1 (b): Fs = 200 Hz → x(n) = 3 cos(πn/2)



Example 1.1 (c, d): Fs = 75 Hz → same samples as 25 Hz



Final 23-1-2023 (نفس النوع: Fs كافي ولا لأ + x(n)) Final 11-6-2023 Final 20-1-2025 جه في

P2 Example 1.2 — Nyquist rate

Consider $x_a(t) = 3 \cos 50\pi t + 10 \sin 300\pi t - \cos 100\pi t$. What is the Nyquist rate for this signal?

SOLUTION

The frequencies present in the signal above are f1 = 25 Hz, f2 = 150 Hz, f3 = 50 Hz. Thus fmax = 150 Hz and according to (1.47), Fs > 2fmax = 300 Hz. The Nyquist rate is fN = 2fmax. Hence fN = 300 Hz.

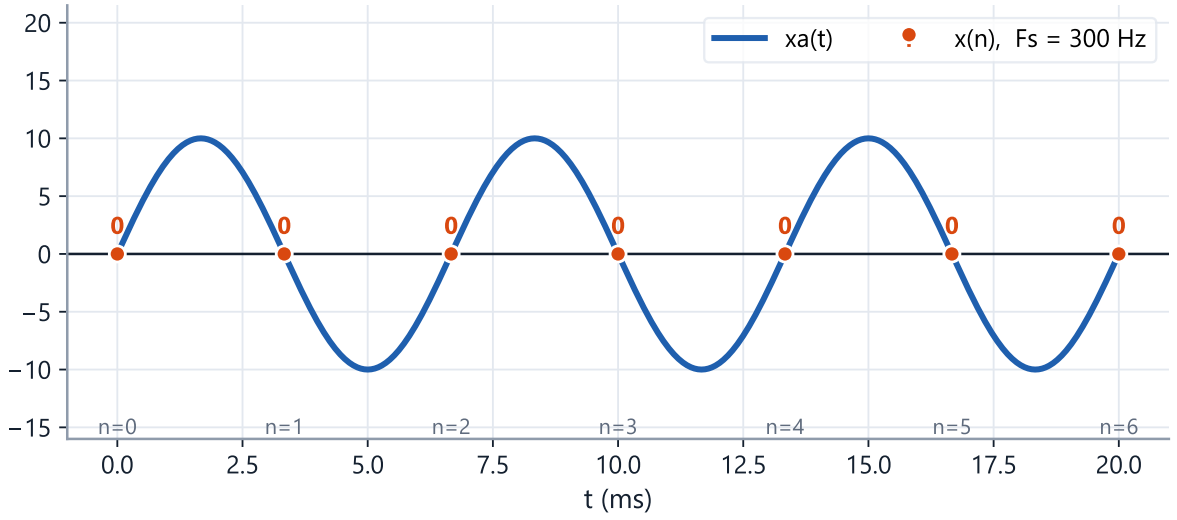
Discussion

- It should be observed that the signal component 10 sin 300πt, sampled at the Nyquist rate fN = 300, results in the samples 10 sin πn, which are identically zero.
- In other words, we are sampling the analog sinusoid at its zero-crossing points, and hence we miss this signal component completely.
- This situation would not occur if the sinusoid is offset in phase by some amount θ. In such a case we have 10 sin(300πt + θ) sampled at the Nyquist rate fN = 300 samples per second, which yields the samples:

$$10 \sin(\pi n + \theta) = 10(\sin \pi n \cos \theta + \cos \pi n \sin \theta) = 10 \sin \theta \cos \pi n = (-1)^n 10 \sin \theta$$

- Thus if $\theta \neq 0$ or π, the samples of the sinusoid taken at the Nyquist rate are not all zero.
- However, we still cannot obtain the correct amplitude from the samples when the phase θ is unknown.
- A simple remedy that avoids this potentially troublesome situation is to sample the analog signal at a rate higher than the Nyquist rate.

Example 1.2: 10 sin 300πt at Fs = 300 Hz → all samples are zero



جه في Final 12-1-2019 Final 28-12-2019 كل امتحان فيه سؤال Nyquist

P3 Example 1.3 — Aliasing with three components

Consider $x_a(t) = 3 \cos 2000\pi t + 5 \sin 6000\pi t + 10 \cos 12000\pi t$.

a) What is the Nyquist rate?

b) The signal is sampled at Fs = 5000 samples/s. What is the discrete-time signal?

c) What is the analog signal $y_a(t)$ reconstructed using ideal interpolation?

SOLUTION

(a) The frequencies existing in the analog signal are f1 = 1 kHz, f2 = 3 kHz, f3 = 6 kHz. Thus fmax = 6 kHz, and according to the sampling theorem, Fs > 2fmax = 12 kHz. The Nyquist rate is fN = 12 kHz.

(b) Since we have chosen Fs = 5 kHz, the folding frequency is Fs/2 = 2.5 kHz, and this is the maximum frequency that can be represented uniquely by the sampled signal. By making use of (1.30) we obtain:

$$\begin{aligned} x(n) &= xa(nT) = xa(n/Fs) \\ &= 3 \cos 2\pi(1/5)n + 5 \sin 2\pi(3/5)n + 10 \cos 2\pi(6/5)n \\ &= 3 \cos 2\pi(1/5)n + 5 \sin 2\pi(1 - 2/5)n + 10 \cos 2\pi(1 + 1/5)n \\ &= 3 \cos 2\pi(1/5)n + 5 \sin 2\pi(-2/5)n + 10 \cos 2\pi(1/5)n \end{aligned}$$

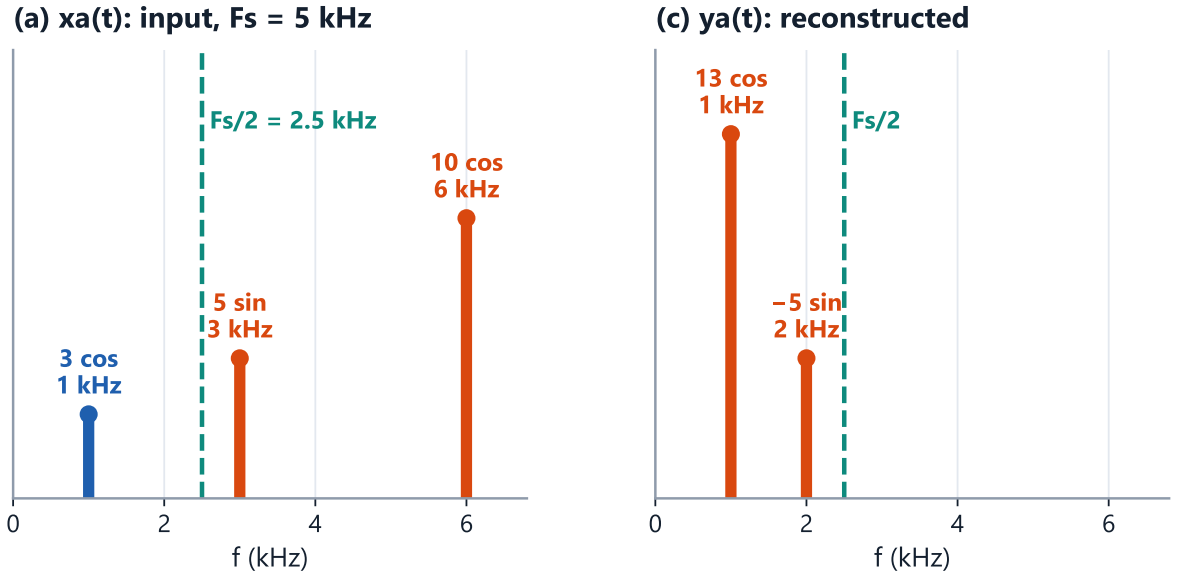
Finally, we obtain:

$$x(n) = 13 \cos 2\pi(1/5)n - 5 \sin 2\pi(2/5)n$$

(c) Since only the frequency components at 1 kHz and 2 kHz are present in the sampled signal, the analog signal we can recover is

$$ya(t) = 13 \cos 2000\pi t - 5 \sin 4000\pi t$$

which is obviously different from the original signal xa(t). This distortion of the original analog signal was caused by the aliasing effect, due to the low sampling rate used.



P4 Example 1.4 — The aliasing concept

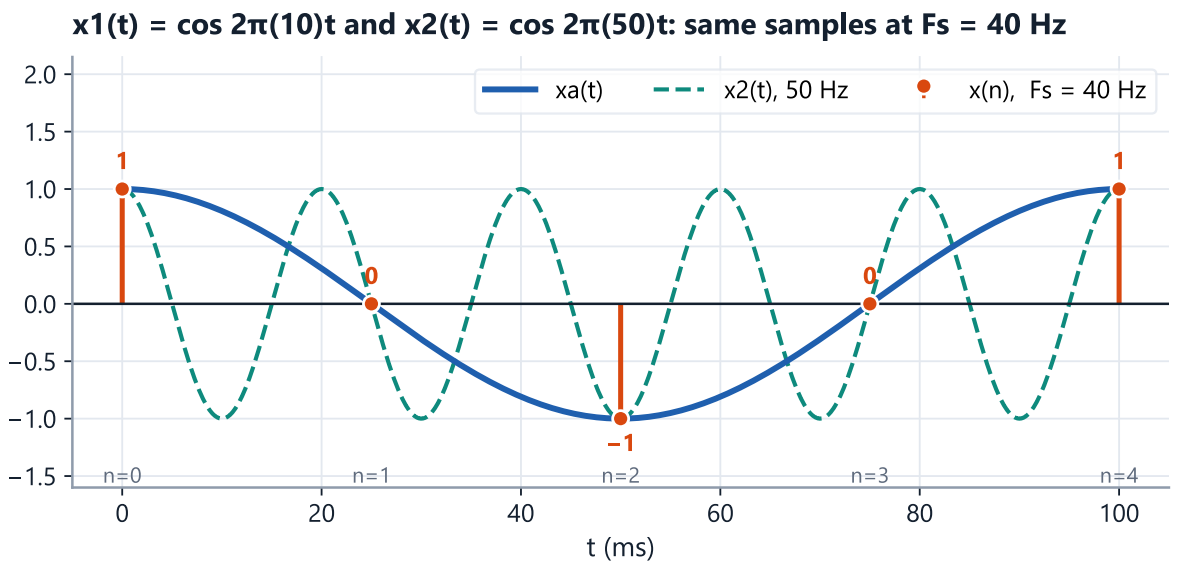
Two signals x1(t) = cos 2π(10)t and x2(t) = cos 2π(50)t are sampled at Fs = 40 Hz. Find x1(n) and x2(n) and discuss.

SOLUTION

The implications of these frequency relations can be fully appreciated by considering the two analog sinusoidal signals x1(t) = cos 2π(10)t and x2(t) = cos 2π(50)t, which are sampled at a rate Fs = 40 Hz. The corresponding discrete-time signals or sequences are:

$$x1(n) = \cos 2\pi(10/40)n = \cos(\pi/2)n$$
$$x2(n) = \cos 2\pi(50/40)n = \cos(5\pi/2)n$$

- However, cos 5πn/2 = cos(2πn + πn/2) = cos πn/2. Hence x2(n) = x1(n).
- Thus the sinusoidal signals are identical and, consequently, indistinguishable.
- If we are given the sampled values generated by cos(π/2)n, there is some ambiguity as to whether these sampled values correspond to x1(t) or x2(t).
- Since x2(t) yields exactly the same values as x1(t) when the two are sampled at Fs = 40 samples per second, we say that the frequency f2 = 50 Hz is an alias of the frequency f1 = 10 Hz at the sampling rate of 40 samples per second.
- It is important to note that f2 is not the only alias of f1. In fact at the sampling rate of 40 samples per second, the frequency f3 = 90 Hz is also an alias of f1, as is the frequency f4 = 130 Hz, and so on.



P5 Quantization example (Table 1.2)

$x_a(t) = 0.9^t$, $t \geq 0$ is sampled at Fs = 1 Hz, giving $x(n) = 0.9^n$. Quantize the first 10 samples to one significant digit by truncation and by rounding; find the error, Δ, and the number of bits.

SOLUTION

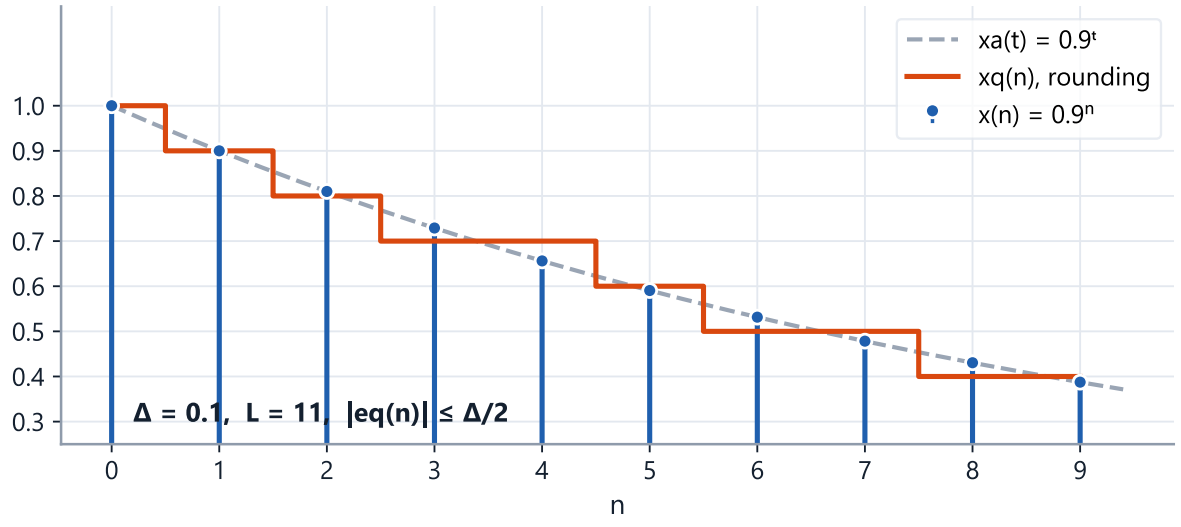
- Let us consider the discrete-time signal $x(n) = 0.9^n$ for $n \geq 0$ (0 for $n < 0$), obtained by sampling the analog exponential signal $xa(t) = 0.9^t$, $t \geq 0$ with a sampling frequency Fs = 1 Hz (Ts = 1 sec), see Fig. 1.15(a).
- Observation of Table 1.2, which shows the values of the first 10 samples of x(n), reveals that the description of the sample value x(n) requires n significant digits.
- However, let us assume that we want to use only one significant digit. To eliminate the excess digits, we can either simply discard them (truncation) or discard them by rounding the resulting number (rounding).
- The resulting quantized signals xq(n) are shown in Table 1.2. We discuss only quantization by rounding, although it is just as easy to treat truncation. The rounding process is graphically illustrated in Fig. 1.15b.
- In our example we have xmax = 1, xmin = 0, and L = 11, which leads to Δ = 0.1.
- In our example it can easily be seen that we need a coder with b = 4 bits.

n	x(n)	Truncation	Rounding	eq(n)
0	1	1.0	1.0	0.0
1	0.9	0.9	0.9	0.0
2	0.81	0.8	0.8	−0.01
3	0.729	0.7	0.7	−0.029
4	0.6561	0.6	0.7	0.0439
5	0.59049	0.5	0.6	0.00951
6	0.531441	0.5	0.5	−0.031441
7	0.4782969	0.4	0.5	0.0217031
8	0.43046721	0.4	0.4	−0.03046721
9	0.387420489	0.3	0.4	0.012579511

$\Delta = (x_{max} - x_{min})/(L - 1) = (1 - 0)/(11 - 1) = \mathbf{0.1}$ $e_q(n) = x_q(n) - x(n)$, $|e_q(n)| \le \Delta/2 = 0.05$

$2^b \ge L = 11 \rightarrow \mathbf{b = 4\ bits}$

Quantization with one significant digit (Table 1.2)



P6 Periodicity examples (Lec 3)

a) Find the fundamental period of a DT sinusoid with F1 = 31/60 and F2 = 30/60.

b) For x(n) = cos $\Omega_0 n$, find N for Ω_0 = 0, π/8, π/4, π/2, π.

c) Fig. 1.7: Ω = π/6. Find F and N.

SOLUTION

F = k/N (1.16): cancel common factors so that k and N are relatively prime → N is the

fundamental period

a) F1 = 31/60 → k = 31, N = 60 (relatively prime) → **N1 = 60**. F2 = 30/60 = 1/2 → **N2 = 2**. A small change in frequency results in a large change in the period.

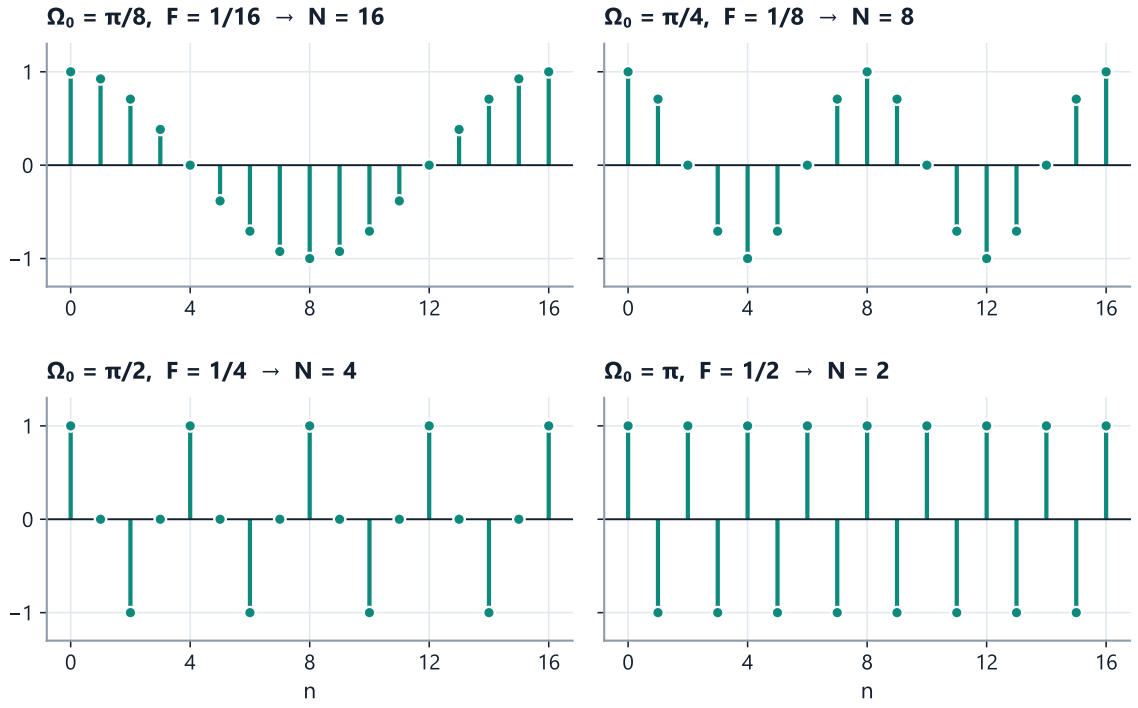
b) $x(n) = \cos \Omega_0 n$, $F = \Omega_0/2\pi$:

$\Omega_0 = 0 \rightarrow F = 0 \rightarrow N = \infty$ | $\pi/8 \rightarrow 1/16 \rightarrow N = 16$ | $\pi/4 \rightarrow 1/8 \rightarrow N = 8$

$\pi/2 \rightarrow 1/4 \rightarrow N = 4$ | $\pi \rightarrow 1/2 \rightarrow N = 2$

The period decreases as the frequency increases (highest rate of oscillation at $\Omega = \pi$).

c) $\Omega = \pi/6 \rightarrow F = (\pi/6)/2\pi = 1/12 \rightarrow$ **N = 12**



SECTION 05

مسائل الامتحانات القديمة

1. Periodicity of continuous-time and discrete-time signals
2. $x(t) = \cos(200\pi t)$, $F_s = 400$ Hz
3. $x(t) = 10\cos(1500\pi t)$, $F_s = 3000$ samples/s
4. $x(t) = 10\cos(1000\pi t)$, $F_s = 2000$ sample/s
5. $x(t) = A\cos(2\pi f_0 t)$, $A = 2$ V, $f_0 = 10$ kHz
6. $x(t) = 2\cos(2000\pi t)$, $F_s = 3000$ Hz then $F_s = 8000$ Hz
7. $x(t) = 3\sin(100\pi t)$, $F_s = 300$ samples/s
8. Is F_s suitable? (F_s = Nyquist rate exactly)
9. Two signals with two sampling frequencies
10. A/D then D/A with different sampling periods (Sheet 1.8)

P7 Periodicity of continuous-time and discrete-time signals

متكرر في الامتحانات

Check the periodicity of the following signals and find their fundamental periods:
(i) $x_a(t) = 3\cos(5t + \pi/6)$ (ii) $x(n) = \cos(3\pi n)$ (iii) $x(n) = \cos(0.01\pi n)$ (iv) $X(n) = 2\sin(3\pi n/105)$ (v) $x(t) = \cos(3000\pi t)$ (vi) $y(t) = 7\cos(400\pi t)$ (vii) $x(n) = \sin(2n + \pi/4)$

SOLUTION

(i)

xa(t) = 3 cos(5t + π/6) is a continuous-time sinusoid xa(t) = A cos(2πft + θ). For every fixed value of f, xa(t) is periodic with Tp = 1/f.

$$2\pi f = 5 \rightarrow f = 5/2\pi \text{ Hz} \rightarrow \mathbf{Tp = 1/f = 2\pi/5 \text{ sec}}$$
$$x_a(t + Tp) = 3\cos(5(t + 2\pi/5) + \pi/6) = 3\cos(5t + 2\pi + \pi/6) = x_a(t)$$

∴ Periodic, fundamental period Tp = 2π/5 sec.

(ii)

x(n) = cos(3πn) is a discrete-time sinusoid x(n) = A cos(2πFn + θ): it is periodic only if F is rational (F = k/N).

$$2\pi F = 3\pi \rightarrow F = 3/2 = k/N \text{ (k and N relatively prime)} \rightarrow \text{rational}$$
$$x(n + 2) = \cos(3\pi n + 6\pi) = \cos(3\pi n) = x(n)$$

∴ Periodic, fundamental period **N = 2 samples**.

(iii)

x(n) = cos(0.01πn) is a discrete-time sinusoid x(n) = A cos(2πFn + θ): it is periodic only if F is rational (F = k/N).

$$2\pi F = 0.01\pi \rightarrow F = 1/200 = k/N \text{ (k and N relatively prime)} \rightarrow \text{rational}$$
$$x(n + 200) = \cos(0.01\pi n + 2\pi) = x(n)$$

∴ Periodic, fundamental period **N = 200 samples**.

(iv)

x(n) = 2 sin(3πn/105) is a discrete-time sinusoid x(n) = A cos(2πFn + θ): it is periodic only if F is rational (F = k/N).

$$2\pi F = 3\pi/105 = \pi/35 \rightarrow F = 1/70 = k/N \text{ (k and N relatively prime)} \rightarrow \text{rational}$$
$$x(n + 70) = 2\sin(\pi n/35 + 2\pi) = x(n)$$

∴ Periodic, fundamental period **N = 70 samples**.

(v)

x(t) = cos(3000πt) is a continuous-time sinusoid xa(t) = A cos(2πft + θ). For every fixed value of f, xa(t) is periodic with Tp = 1/f.

$$2\pi f = 3000\pi \rightarrow f = 1500 \text{ Hz} \rightarrow \mathbf{Tp = 1/f = 1/1500 \text{ sec} = 0.667 \text{ msec}}$$
$$x(t + Tp) = \cos(3000\pi t + 2\pi) = x(t)$$

∴ Periodic, fundamental period Tp = 1/1500 sec = 0.667 msec.

(vi)

y(t) = 7 cos(400πt) is a continuous-time sinusoid xa(t) = A cos(2πft + θ). For every fixed value of f, xa(t) is periodic with Tp = 1/f.

$$2\pi f = 400\pi \rightarrow f = 200 \text{ Hz} \rightarrow \mathbf{Tp = 1/f = 5 \text{ msec}}$$
$$y(t + Tp) = 7\cos(400\pi t + 2\pi) = y(t)$$

∴ Periodic, fundamental period Tp = 5 msec.

(vii)

x(n) = sin(2n + π/4) is a discrete-time sinusoid x(n) = A cos(2πFn + θ): it is periodic only if F is rational (F = k/N).

$$2\pi F = 2 \rightarrow F = 2/2\pi = 1/\pi$$

F cannot be expressed as a ratio of two integers → **not periodic**.

Midterm 8-12-2020 | Midterm 1 Model-2 | Midterm 31-7-2023 | Midterm 26-7-2026 | جه في
Midterm OLD | Final 2-1-2017 | Final 23-12-2017 | Final 12-1-2019 | Final 27-6-2021

P8 x(t) = cos(200πt), Fs = 400 Hz

متكرر في الامتحانات

For the signal x(t) = cos(200πt) to be sampled with Fs = 400 Hz:
(i) Find an expression for its corresponding sequence x(n). Then calculate its fundamental period N_p.
(ii) Find the sample values of the first cycle in x(n), then plot x(n) and x(t) on the same figure.

SOLUTION

$$x(t) = 1\cos(200\pi t) = A\cos(2\pi f t + \theta) \rightarrow A = 1, \quad 2\pi f = 200\pi \rightarrow \mathbf{f = 100 \text{ Hz}}$$

$$F_N = 2f_{max} = 2 \times 100 = \mathbf{200 \text{ Hz}}$$
$$Fs = 400 \text{ Hz} \geq 2f_{max} = 200 \text{ Hz}$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

(i)

If the signal is sampled at Fs = 400 Hz (t = nT = n/Fs), the discrete-time signal is:

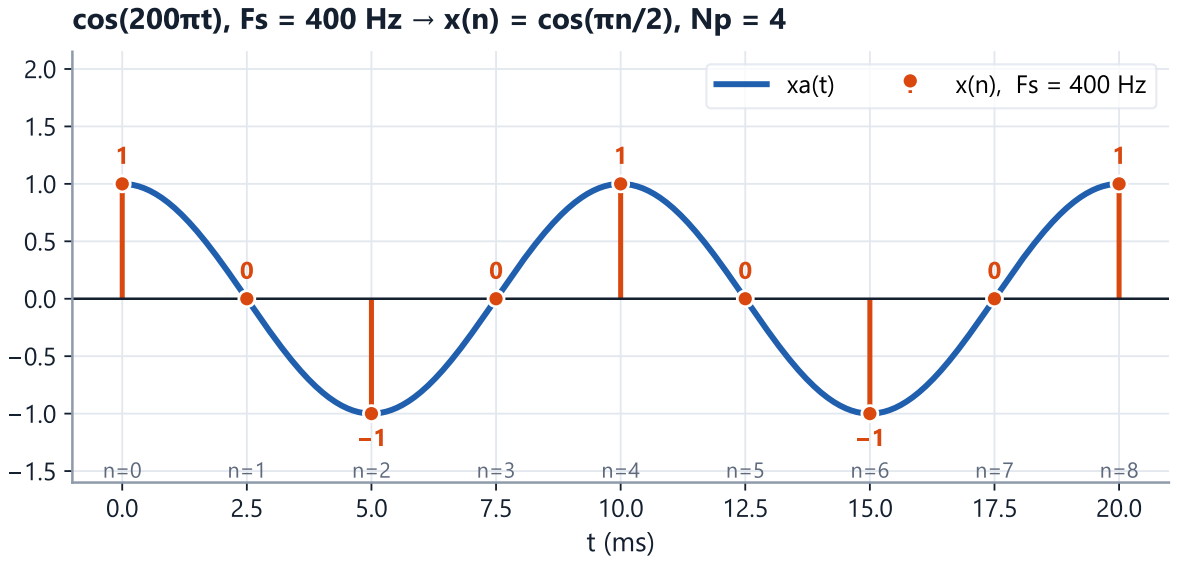
$$x(n) = xa(nT) = xa(n/Fs) = 1\cos(200\pi\, n/400) = \mathbf{1\cos(\pi/2\, n)}$$

$$F = f/Fs = 100/400 = \mathbf{1/4} = k/N_P \;\rightarrow\; \text{rational} \rightarrow \text{periodic}, \quad \mathbf{N_P = 4\ samples}$$

The sample values in the first cycle (n = 0, 1, ..., 3):

$$x(0) = 1\cos(0) = \mathbf{1}$$
$$x(1) = 1\cos(\pi/2) = \mathbf{0}$$
$$x(2) = 1\cos(\pi) = \mathbf{-1}$$
$$x(3) = 1\cos(3\pi/2) = \mathbf{0}$$

(ii) Plot x(t) and x(n) on the same figure:



Final 7-6-2023 (Q4-b) | Midterm 1 Model-2 | Midterm 31-7-2023 | Midterm 26-7-2026 | جه في

Midterm OLD (أكتر من نموذج) | Quiz 1 (Section) | Final 18-1-2024 | Final 8-9-2021

P9 x(t) = 10 cos(1500πt), Fs = 3000 samples/s

For x(t) = 10 cos(1500πt):

(i) If it is sampled with Fs = 3000 sample/s, is it possible to reconstruct this signal from its samples? Find the Nyquist frequency F_N.

(ii) Using Fs, find x(n), then calculate the sample values during one cycle and plot x(t) and x(n).

SOLUTION

$$x(t) = 10\cos(1500\pi t) = A\cos(2\pi ft + \theta) \;\rightarrow\; A = 10, \quad 2\pi f = 1500\pi \;\rightarrow\; \mathbf{f = 750\ Hz}$$

(i), (ii)

$$F_N = 2f_{max} = 2 \times 750 = \mathbf{1500\ Hz}$$
$$Fs = 3000\ Hz \geq 2f_{max} = 1500\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

(iii), (iv)

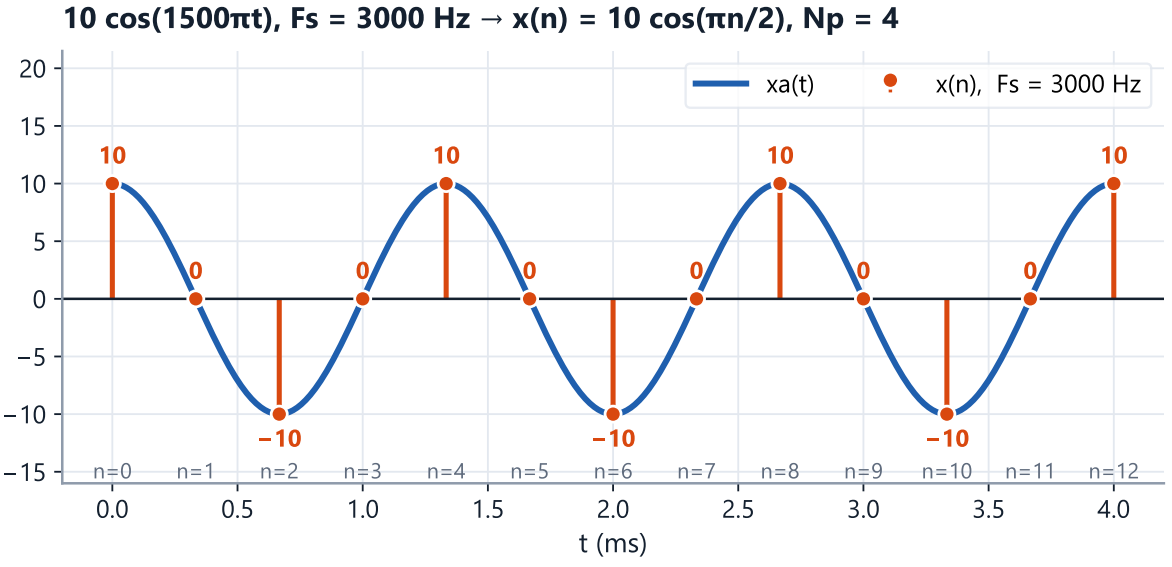
If the signal is sampled at Fs = 3000 Hz (t = nT = n/Fs), the discrete-time signal is:

$$x(n) = xa(nT) = xa(n/Fs) = 10\cos(1500\pi\, n/3000) = \mathbf{10\cos(\pi/2\, n)}$$

$$F = f/Fs = 750/3000 = \mathbf{1/4} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic}, \quad \mathbf{N_P = 4\ samples}$$

The sample values in the first cycle (n = 0, 1, ..., 3):

$$x(0) = 10\cos(0) = \mathbf{10}$$
$$x(1) = 10\cos(\pi/2) = \mathbf{0}$$
$$x(2) = 10\cos(\pi) = \mathbf{-10}$$
$$x(3) = 10\cos(3\pi/2) = \mathbf{0}$$



Revision Sheet 1 (Q9) Final 28-12-2019 Final 12-1-2019 جه في

P10 x(t) = 10 cos(1000πt), Fs = 2000 sample/s

For x(t) = 10 cos(1000πt):

(i) If sampled with Fs = 2000 sample/s, is it possible to reconstruct it successfully?

(ii) Find the Nyquist frequency F_N.

(iii) Using F_N, find an expression for x(n).

(iv) Find the values of x(n) in the first cycle.

SOLUTION

$$x(t) = 10\cos(1000\pi t) = A\cos(2\pi f t + \theta) \rightarrow A = 10,\quad 2\pi f = 1000\pi \rightarrow \mathbf{f = 500\ Hz}$$

(i), (ii)

$$F_N = 2f_{max} = 2 \times 500 = \mathbf{1000\ Hz}$$
$$Fs = 2000\ Hz \geq 2f_{max} = 1000\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

(iii) Using the obtained Nyquist frequency F_N = 1000 Hz:

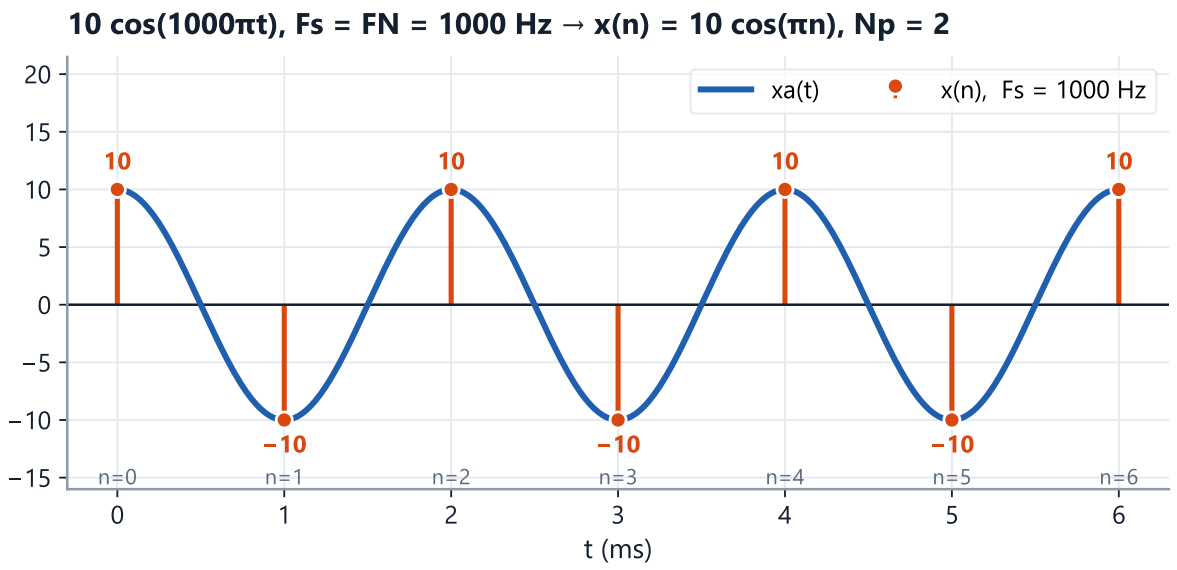
If the signal is sampled at F_N = 1000 Hz (t = nT = n/Fs), the discrete-time signal is:

$$x(n) = xa(nT) = xa(n/Fs) = 10\cos(1000\pi\, n/1000) = \mathbf{10\cos(\pi\, n)}$$

$$F = f/Fs = 500/1000 = \mathbf{1/2} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic},\quad \mathbf{N_P = 2\ samples}$$

The sample values in the first cycle (n = 0, 1, ..., 1):

$$x(0) = 10\cos(0) = \mathbf{10}$$
$$x(1) = 10\cos(\pi) = \mathbf{-10}$$



Final 23-1-2023 (Q4-b) Final 20-1-2025 (Q3-c) Final 11-6-2023 (Q1-b) جه في

P11 x(t) = A cos(2πf₀t), A = 2 V, f₀ = 10 kHz

متكرر في الامتحانات

For $x_a(t) = A\cos(2\pi f_0 t)$, A = 2 volt, f_0 = 10 kHz:

a) Check the periodicity and find the fundamental period.

b) If sampled at Fs = 5 kHz, is it possible to reconstruct the original signal?

c) What is the minimum sampling frequency?

d) Using it, find x(n) and the sample values during one cycle (or the first four samples).

SOLUTION

$$xa(t) = A\cos(2\pi f_0 t),\ A = 2\ V,\ f_0 = 10\ kHz \rightarrow xa(t) = 2\cos(20000\pi t)$$

a) For every fixed value of f, xa(t) is periodic: **$Tp = 1/f_0 = 1/10^4 = 0.1$ msec**.

b)

$$F_N = 2f_{max} = 2 \times 10 = \mathbf{20\ kHz}$$
$$Fs = 5\ kHz < 2f_{max} = 20\ kHz$$

Since $Fs < 2f_{max}$, aliasing occurs and the signal **cannot be reconstructed** from its samples.

c) The minimum sampling frequency is **$Fs = 2f_{max} = 20$ kHz**.

d)

If the signal is sampled at Fs = 20000 Hz ($t = nT = n/Fs$), the discrete-time signal is:

$$x(n) = xa(nT) = xa(n/Fs) = 2\cos(20000\pi\ n/20000) = \mathbf{2\cos(\pi\ n)}$$

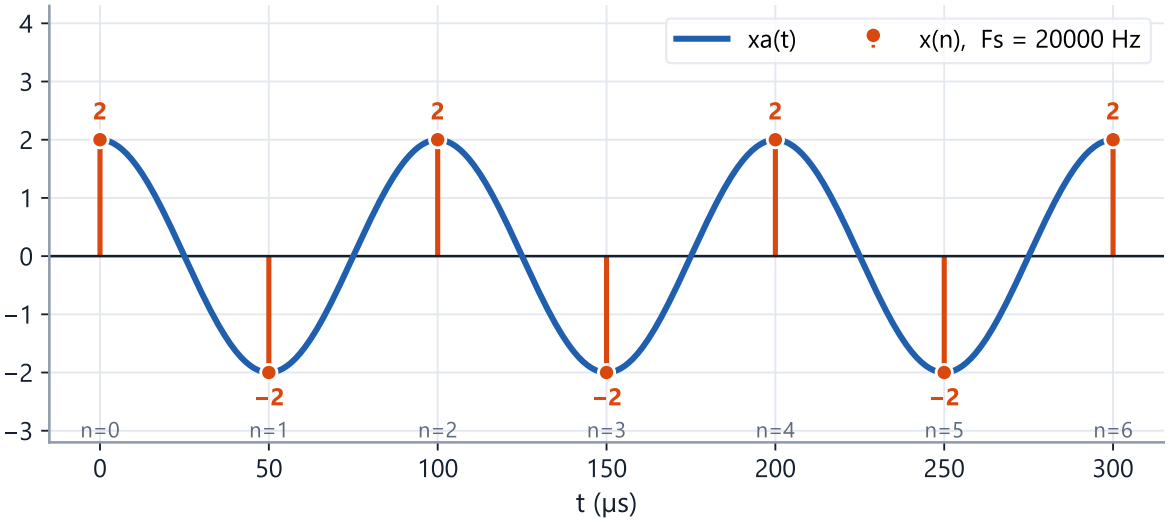
$$F = f/Fs = 10000/20000 = \mathbf{1/2} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic}, \quad \mathbf{N_P = 2 \text{ samples}}$$

The sample values in the first cycle (n = 0, 1, ..., 1):

$$x(0) = 2\cos(0) = \mathbf{2}$$
$$x(1) = 2\cos(\pi) = \mathbf{-2}$$

First four samples: x(0) = 2, x(1) = −2, x(2) = 2, x(3) = −2

2 cos(2π·10k·t), Fs = 20 kHz → x(n) = 2 cos(πn), Np = 2



Revision Sheet 1 (Q10) Quiz 1 Summer 26 (Q2) Final 17-8-2026 (Q1-b) Final 9-6-2026 (Q1-b) جه في

P12 x(t) = 2 cos(2000πt), Fs = 3000 Hz then Fs = 8000 Hz

Given x(t) = 2 cos(2000πt) to be sampled with Fs = 3000 Hz:
(i) Could the original signal be reconstructed successfully from its samples?
(ii) Take Fs = 8000 Hz; find x(n) and its fundamental period N_P.
(iii) Find the sample values of the first cycle and plot x(n).

SOLUTION

$$x(t) = 2\cos(2000\pi t) = A\cos(2\pi ft + \theta) \rightarrow A = 2, \quad 2\pi f = 2000\pi \rightarrow \mathbf{f = 1000 \text{ Hz}}$$

(i)

$$F_N = 2f_{max} = 2 \times 1000 = \mathbf{2000 \text{ Hz}}$$
$$Fs = 3000 \text{ Hz} \geq 2f_{max} = 2000 \text{ Hz}$$

Since Fs ≥ $2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

(ii), (iii)

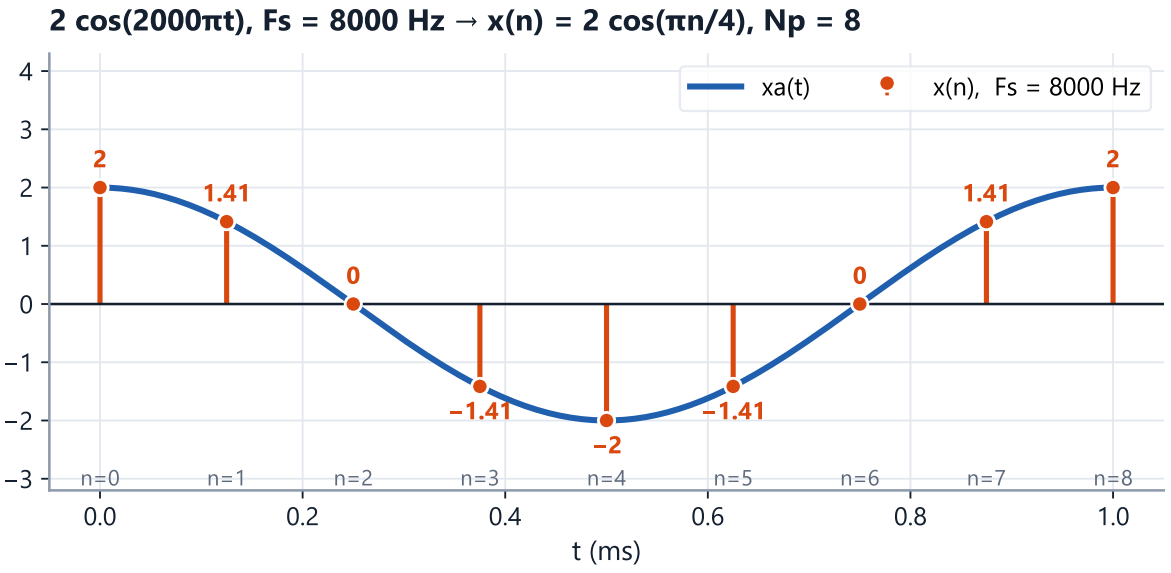
If the signal is sampled at Fs = 8000 Hz (t = nT = n/Fs), the discrete-time signal is:

$$x(n) = x_a(nT) = x_a(n/Fs) = 2\cos(2000\pi\, n/8000) = \mathbf{2\cos(\pi/4\, n)}$$

$$F = f/Fs = 1000/8000 = \mathbf{1/8} = k/N_P \;\rightarrow\; \text{rational} \rightarrow \text{periodic}, \quad \mathbf{N_P = 8 \text{ samples}}$$

The sample values in the first cycle (n = 0, 1, …, 7):

$$x(0) = 2\cos(0) = \mathbf{2}$$
$$x(1) = 2\cos(\pi/4) = \mathbf{1.414}$$
$$x(2) = 2\cos(\pi/2) = \mathbf{0}$$
$$x(3) = 2\cos(3\pi/4) = \mathbf{-1.414}$$
$$x(4) = 2\cos(\pi) = \mathbf{-2}$$
$$x(5) = 2\cos(5\pi/4) = \mathbf{-1.414}$$
$$x(6) = 2\cos(3\pi/2) = \mathbf{0}$$
$$x(7) = 2\cos(7\pi/4) = \mathbf{1.414}$$



Midterm Solve 14-11-2019 (Q1-c) Midterm 12-11-2025 (Q1) جه في

P13 x(t) = 3 sin(100πt), Fs = 300 samples/s

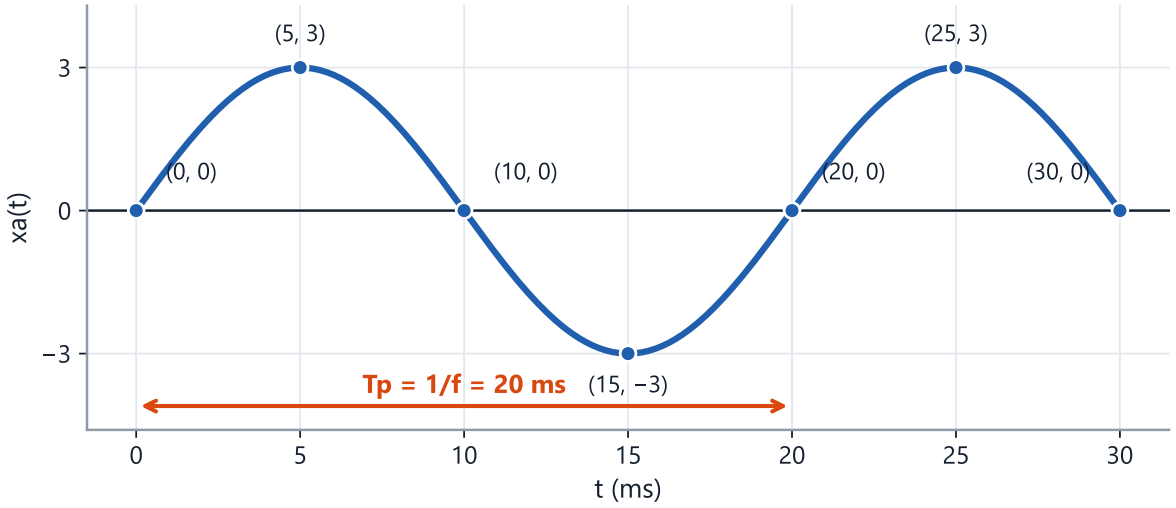
متكرر في الامتحانات

Consider x(t) = 3 sin(100πt):
(i) Sketch the signal for [0, 30] ms.
(ii) The signal is sampled at Fs = 300 samples/s. Determine the frequency of x(n) and show that it is periodic.
(iii) Compute the sample values in one period of x(n). Sketch x(n) on the same diagram with x(t). What is the period of x(n)?
(iv) Can you find Fs such that x(n) reaches its peak value 3? What is the minimum Fs?

SOLUTION

(i) xa(t) = 3 sin(100πt): A = 3, 2πf = 100π → f = 50 Hz, Tp = 1/f = 20 msec. The interval [0, 30] ms contains 1.5 periods.

1.7 (a) xa(t) = 3 sin(100πt) on [0, 30] ms (f = 50 Hz, 1.5 cycles)



(ii), (iii)

If the signal is sampled at Fs = 300 Hz (t = nT = n/Fs), the discrete-time signal is:

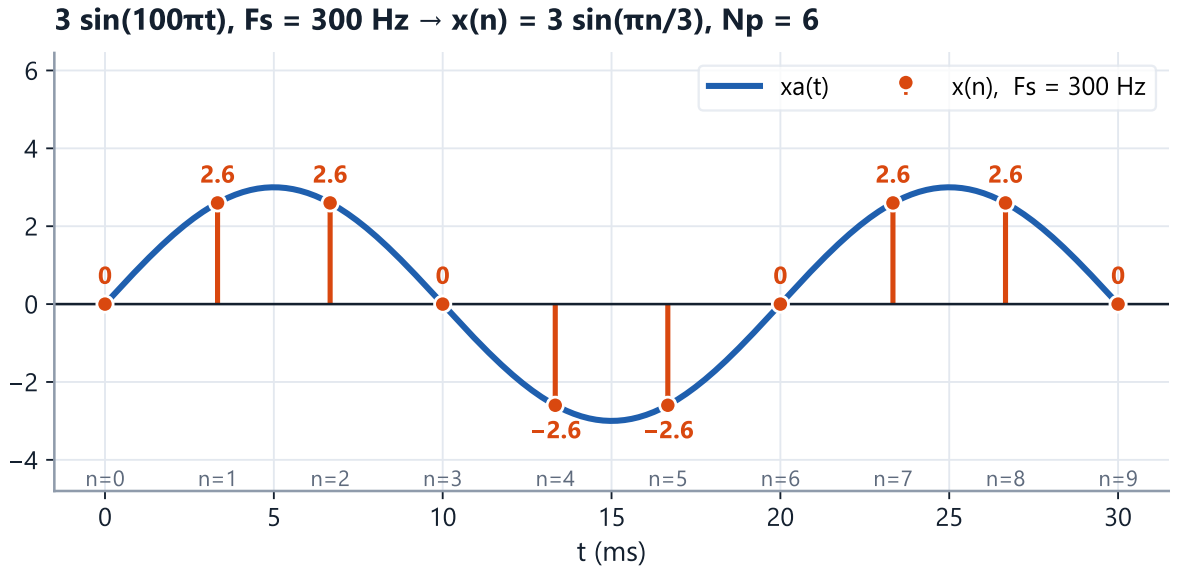
$$x(n) = x_a(nT) = x_a(n/F_s) = 3\sin(100\pi\, n/300) = \mathbf{3\sin(\pi/3\, n)}$$

$$F = f/F_s = 50/300 = \mathbf{1/6} = k/N_P \;\rightarrow\; \text{rational} \rightarrow \text{periodic}, \quad \mathbf{N_P = 6 \text{ samples}}$$

The sample values in the first cycle (n = 0, 1, ..., 5):

$$x(0) = 3\sin(0) = \mathbf{0}$$
$$x(1) = 3\sin(\pi/3) = \mathbf{2.598}$$
$$x(2) = 3\sin(2\pi/3) = \mathbf{2.598}$$
$$x(3) = 3\sin(\pi) = \mathbf{0}$$
$$x(4) = 3\sin(4\pi/3) = \mathbf{-2.598}$$
$$x(5) = 3\sin(5\pi/3) = \mathbf{-2.598}$$

Period of x(n): N = 6 samples = 6 × (1/300) s = 20 msec.



(iv) x(n) = 3 sin(100πn/Fs) reaches its peak value 3 when:

$$\sin(100\pi n/Fs) = 1 \rightarrow 100\pi n/Fs = \pi/2 \rightarrow Fs = 200n \rightarrow \text{minimum at } n = 1: \mathbf{Fs = 200 \text{ samples/sec}}$$

Final 4-3-2021 (Q1-b) Midterm 8-12-2020 (Q1-c) Midterm 2 (Q1-c) Midterm 28-11-2021 (Q1-b) جه في Sheet 1 (1.7)

P14 Is Fs suitable? (Fs = Nyquist rate exactly)

(a) x(t) = 2 cos(3000πt), Fs = 3 kHz. (b) x(t) = 2 cos(2000πt), Fs = 2 kHz. (c) x(t) = cos(300πt), Fs = 300 Hz.

Is Fs suitable? If so, find x(n), the values in the first cycle, and plot.

SOLUTION

(a)

$$x(t) = 2\cos(3000\pi t) = A\cos(2\pi ft + \theta) \rightarrow A = 2, \quad 2\pi f = 3000\pi \rightarrow \mathbf{f = 1500 \text{ Hz}}$$

$$F_N = 2f_{max} = 2 \times 1500 = \mathbf{3000 \text{ Hz}}$$
$$Fs = 3000 \text{ Hz} \geq 2f_{max} = 3000 \text{ Hz}$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

Fs = $2f_{max}$ exactly (the Nyquist rate): it is the minimum rate. Sampling at a rate higher than the Nyquist rate is preferred (Example 1.2).

If the signal is sampled at Fs = 3000 Hz (t = nT = n/Fs), the discrete-time signal is:

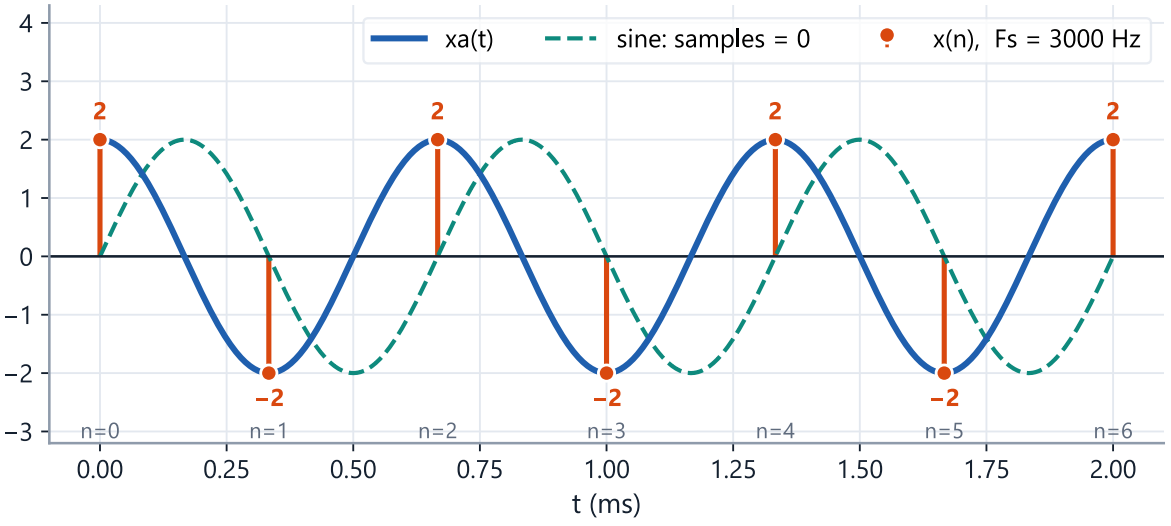
$$x(n) = x_a(nT) = x_a(n/Fs) = 2\cos(3000\pi\, n/3000) = \mathbf{2\cos(\pi\, n)}$$

$$F = f/Fs = 1500/3000 = \mathbf{1/2} = k/N_P \;\rightarrow\; \text{rational} \rightarrow \text{periodic}, \quad \mathbf{N_P = 2\ samples}$$

The sample values in the first cycle (n = 0, 1, ..., 1):

$$x(0) = 2\cos(0) = \mathbf{2}$$
$$x(1) = 2\cos(\pi) = \mathbf{-2}$$

Fs = 2f exactly: x(n) = 2 cos(πn) = 2, −2, ...



(b) x(t) = 2 cos(2000πt), Fs = 2 kHz → f = 1000 Hz, Fs = 2f → x(n) = 2 cos(πn): 2, −2.

(c) x(t) = cos(300πt), Fs = 300 Hz → f = 150 Hz, Fs = 2f → x(n) = cos(πn): 1, −1.

Book Midterm 2015 Model-A | Final 23-12-2017 (Q1-c) | Final 2-1-2017 (Q1-c) | جه في

P15 Two signals with two sampling frequencies

x1(t) = 5 cos 200πt and x2(t) = 2 sin 500πt are sampled with Fs1 = 400 Hz and Fs2 = 1 kHz respectively.

(i) Are the sampling frequencies suitable? (ii) If suitable, find x1(n) and x2(n).

SOLUTION

x1(t):

$$x1(t) = 5\cos(200\pi t) = A\cos(2\pi ft + \theta) \rightarrow A = 5, \quad 2\pi f = 200\pi \rightarrow \mathbf{f = 100\ Hz}$$

$$F_N = 2f_{max} = 2 \times 100 = \mathbf{200\ Hz}$$
$$Fs = 400\ Hz \geq 2f_{max} = 200\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

If the signal is sampled at Fs = 400 Hz (t = nT = n/Fs), the discrete-time signal is:

$$x(n) = xa(nT) = xa(n/Fs) = 5\cos(200\pi\ n/400) = \mathbf{5\cos(\pi/2\ n)}$$

$$F = f/Fs = 100/400 = \mathbf{1/4} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic}, \quad \mathbf{N_P = 4\ samples}$$

x2(t):

$$x2(t) = 2\sin(500\pi t) \rightarrow 2\pi f = 500\pi \rightarrow \mathbf{f = 250\ Hz}$$

$$F_N = 2f_{max} = 2 \times 250 = \mathbf{500\ Hz}$$
$$Fs = 1000\ Hz \geq 2f_{max} = 500\ Hz$$

Since $Fs \geq 2f_{max}$, there is no aliasing and the signal **can be reconstructed successfully** from its samples.

If the signal is sampled at Fs = 1000 Hz (t = nT = n/Fs), the discrete-time signal is:

$$x(n) = xa(nT) = xa(n/Fs) = 2\sin(500\pi\ n/1000) = \mathbf{2\sin(\pi/2\ n)}$$

$$F = f/Fs = 250/1000 = \mathbf{1/4} = k/N_P \rightarrow \text{rational} \rightarrow \text{periodic}, \quad \mathbf{N_P = 4\ samples}$$

∴ Both sampling frequencies are suitable.

Book Midterm 2015 Model-B جه في

P16 A/D then D/A with different sampling periods (Sheet 1.8)

A signal processing system: $x_a(t) \rightarrow$ A/D $\rightarrow x(n) \rightarrow$ D/A $\rightarrow y_a(t)$. The sampling periods are T1 = 5 ms (A/D) and T2 = 1 ms (D/A). Determine $y_a(t)$ if $x_a(t) = 3\cos(100\pi t) + 2\sin(250\pi t)$.

SOLUTION

f1 = 50 Hz, f2 = 125 Hz. A/D: Fs = 1/T1 = 200 Hz → folding frequency 100 Hz.

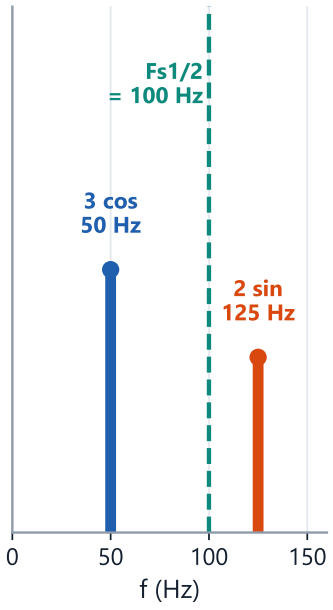
$$x(n) = 3\cos 2\pi(1/4)n + 2\sin 2\pi(5/8)n = 3\cos 2\pi(1/4)n + 2\sin 2\pi(1 - 3/8)n$$

$$\mathbf{x(n) = 3\cos(\pi n/2) - 2\sin(3\pi n/4)}$$

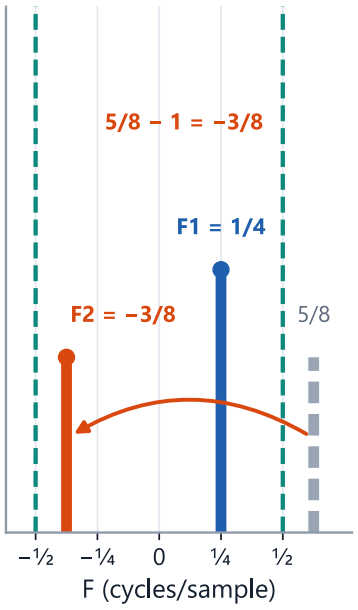
D/A: T2 = 1 ms → n = t/T2 = 1000t:

$$\mathbf{y_a(t) = 3\cos(500\pi t) - 2\sin(750\pi t)}$$

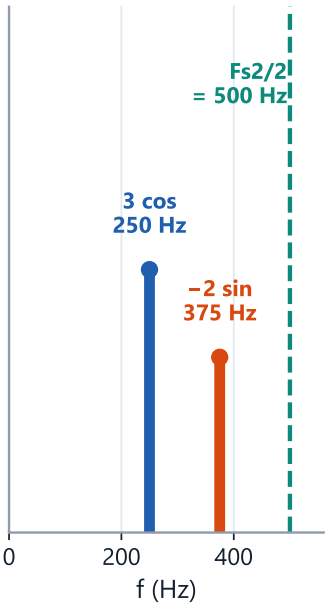
(1) A/D input



(2) x(n): F in ±½



(3) D/A output



Sheet 1 (1.8) Revision Sheet 1 (Q12) جه في